
EDUCATION:

Bachelor of Engineering**Cairo University****2020 - 2025****Computer and Communication Department Track -C (Credit hours system) GPA 3.63**

EXPERIENCE:

- Backend Intern*Mazeed IT — July 2023 - August 2023*

Worked on RESTful web services and their architecture, HTTP protocol and its methods, Routing and controllers in .NET Web API, Serialization and deserialization of data, Authentication and authorization.

- Software Engineer Intern*Incrypt Solutions — August 2024 – November 2024*

Collaborated on production-grade applications, focusing on customer-centric solutions. Utilized technologies such as TypeScript, Next.js, and Node.js for full-stack development. Integrated and managed various third-party APIs to enhance functionality and user experience.

- Software Engineer Trainee*Siemens Digital Industries Software — July 2024 – December 2024*

Engaged in comprehensive training covering key topics: Introduction to Algorithms & Problem Solving, Tools and Utilities, Software Design, and Effective Coding Practices. Gained practical knowledge of the Software Development Life Cycle (SDLC), enhancing understanding of industry-standard processes. Contributed to automating the SW Academy Management System by streamlining key processes such as user management and attendance tracking, significantly improving operational efficiency and system scalability.

PROJECTS:

- Conversational AI for Alzheimer's Patients:

- Engineered a multi-model conversational AI platform integrating advanced NLP capabilities (NER, abstractive summarization, question-answering, automated scheduling, and intent classification) to enhance cognitive support and daily functioning for Alzheimer's patients..

Technologies: Python, TensorFlow/PyTorch, BERT, Transformers, spaCy.

- NLP-Powered Pizza Ordering System

- Implemented an intelligent natural language processing system that precisely extracts structured order specifications from unstructured customer inputs, achieving 92% accuracy through sophisticated intent recognition algorithms and contextual entity extraction techniques...

Technologies: Python, TensorFlow, BERT, Seq2Seq models, NLP libraries

- Redditech

- Developed a fully-featured Reddit incorporating a mock server to simulate backend functionalities. The application includes features such as user authentication, post creation, commenting, upvoting/downvoting, and subreddit management.

Technologies : TypeScript, React, Vite, Tailwind CSS, Socket.IO, Git, and Jest

- OS Scheduler with Memory Management

- A simulation project of various scheduling algorithms with simple GUI user interface. The implemented algorithms are round robin, multilevel feedback queue, highest priority first, and shortest job first.

Technologies: C, Make, Git, GD

- Pipeline Processor (5 stages)

- The processor uses Harvard Architecture with one data memory and an instruction memory. Techniques like Data Forwarding and Early Detection of Branch are used for optimization.

- The Algorithm of 1 bit dynamic prediction is implemented and control hazards are solved with minimum cycles loss.

Technologies: VHDL, Git, ModelSim, Quartus Prime

- Real Time Text Editor application

- Developed a collaborative text editor with user registration and authentication, supporting file creation, renaming, deletion, and access control for shared documents (viewer/editor permissions). Implemented real-time collaborative editing with support for bold and italic formatting using CRDT methods to handle concurrency issues, ensuring smooth real-time updates and cursor movements.

Technologies: React, Bootstrap, Java Spring Boot, MongoDB, Socket.IO

- Resurrection Rumble

- It required a lot of math and problem solving and I was the one who is responsible for gun shooting and collision algorithm.

Technologies C++, OpenGL.

- Chess Game using assembly x86:

- Developed real-time multiplayer chess with chat using serial communication and timers. Different computers using serial communication and using timers.

Technologies Assembly x86, serial communication.

- Image processing for detecting balls :

- Applied for a robot project to detect and collect balls, utilizing computer vision techniques for object recognition and localization.

Technologies: Python, OpenCv

TECHNICAL SKILLS:

- Programming Languages: C++, C, Embedded C, JavaScript, C#, Node.js
- Design & Analysis of Algorithms & Proficient in Problem Solving
- Data Structures, Object-Oriented Programming (OOP), Design Patterns
- Experience with Linux, Git
- Familiar with Basic Concepts of Embedded Systems, AVR Microcontrollers, and Driver Implementation

COURSES:

- Supervised Machine Learning: Regression and Classification** from DeepLearning Stanford Coursera.
- Exploratory Data Analysis** for Machine Learning from IBM Coursera

Awards:

Participated in ECPC (Egyptian Collegiate Programming Contest) 2023